

Asuman Celik

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OBJECTIVE	4+ years hands-on experience in Data Science and Machine Learning specialized in Biomedical Informatics research. Experienced in Python, R, Tensorflow, Machine Learning Algorithms. Pursuing a Ph.D. in Computer Science with a focus on AI applications in healthcare. Strong background in Data Science, analytics, and visualization. Looking for data science roles.	
EDUCATION	Stevens Institute of Technology - Hoboken, NJ	Aug 2023 - May 2026
	<i>Ph.D. in Computer Science</i>	
	University of Cincinnati - Cincinnati, OH	Aug 2020 - Aug 2023
	<i>Master of Science in Biomedical Informatics and Information Technology</i>	
TECHNICAL SKILLS	North American University (NAU) - Houston, TX	Aug 2016 - May 2020
	<i>Bachelors of Science in Computer Science, Software Engineering with Minor in Mathematics</i>	
	Overall GPA: 3.51/4.00 - Major GPA: 3.84/4.00	
	• Advanced: Python, R, Git, Linux/Unix. Visual Data Analysis Libraries: NumPy, Pandas, Matplotlib, Scikitlearn, Seaborn, Bioconductor, Circlize • Experienced: Algorithms, C/C++, Tensorflow, PyTorch, Keras, H5py, SQL, MySQL, Django • Moderate: Java, Javascript, PHP, AWS	
EXPERIENCE	Data Science Intern	Sep 2022 - Aug 2023
	<i>Peel9 Inc - Cincinnati, OH</i>	
	• Conducting exploratory data analysis and data preprocessing, including data cleaning, feature engineering, and feature selection. • Developing and implementing machine learning algorithms, including supervised and unsupervised learning methods, such as regression, classification, clustering, and deep learning. • Evaluating and optimizing machine learning models, including using cross-validation, hyperparameter tuning, and other techniques to improve performance. • Tools: Python, Jupyter Notebooks, SQL, Git, PowerBI, Tableau • Supervisor: Said Varlioglu	
	Data Science Researcher	August 2022 - current
	<i>University of Cincinnati</i>	
	• Precision Medicine and GNN applications in drug-drug interactions.	
	<i>Supervisors:</i> Dr. Jess Kropczynski	
	Machine Learning Researcher	August 2021 - May 2022
	<i>Institute of Academic Excellence</i>	
	• Graph Neural Network Approaches for Protein-Drug Interactions	
	<i>Supervisors:</i> Dr. Esra Akbas, Dr. Kubra Karagoz	
	Instructor - Machine Learning and Deep Learning	June 2019 - Aug 2019
	<i>Rice University - Houston, TX</i>	
	• Taught Introduction to ML and AI courses to 30 freshman undergraduate and senior and junior high school students.	

- Mentored and advised students on academic and professional matters, including career paths and research opportunities.
- Courses included:
- Machine Learning and Deep Learning basics
 - Python, NumPy, Matplotlib libraries
 - Tensorflow, Keras, Neural Networks and Convolutional Neural Networks
- Developed more than **12 projects such as Image Detection using Fashion MNIST, and Cifar-10 datasets and face detection using Opencv, Open AI Gym and PSY2.**

TEACHING
EXPERIENCE

Adjunct Professor

University of Cincinnati - Cincinnati, OH

December 2022 - current

- Taught Systems Analysis and Design, an undergraduate level course to 32 students.
- Developed and delivered engaging and informative lectures
- Designed and graded assignments, quizzes, and exams that effectively measured student comprehension and provided feedback to support student success.
- Mentored and advised students on academic and professional matters, including career paths and research opportunities.
- Contributed to the academic community by attending departmental meetings and events, participating in committees, and engaging in ongoing professional development

Graduate Teaching Assistant

University of Cincinnati - Cincinnati, OH

Aug 2022 - current

- Co-teaching four undergraduate level courses for computer science department to over 300 students.
 - Machine Learning (with Python Programming)
 - Data Technologies Programming (Python Programming)
 - Web Server Application Development (SQL, PHP)
 - Computer Programming 1 (Java Programming)
 - Computer Programming 1 (Python Programming)
- Grading as primary responsibility as well as preparing labs and extra sessions using Git/GitHub, Python Programming, and Ipython/Jupyter.

Undergraduate Teaching Assistant

North American University - Houston, TX

Aug 2019 - May 2020

- Co-teaching three undergraduate level courses for computer science department to over 30 students each semester.
 - COMP 1411 Introduction to CS 1(Python Programming)
 - COMP 1412 Introduction to CS 2(Java Programming)
 - COMP 2415 Systems Programming
- Preparing labs and extra sessions on Git/GitHub, Python Programming, and Ipython/Jupyter.
- Sharing responsibilities for preparing lectures, exams, homework assignments, and grading.

Instructor - Python Programming

North American University - Houston, TX

Sep 2017 - May 2018

- Lectured introductory Python programming course to freshman, sophomore and junior students for a semester, and led the Python study group of five students.
- Prepared the course material using Jupyter Lab and Git.
- Assigned and graded weekly Python programming worksheets, lab projects and homeworks.

RESEARCH
EXPERIENCE

Cleveland Clinic Lerner Research Center

Visiting Researcher

June 2022 - August 2022

- Analysis of Differentially Methylated Regions of whole genome data using R, "circlize" package.
Supervisors: Dr. Lynn Bekris

Cincinnati Children's Hospital Medical Center **Cincinnati, OH**
Graduate Research Assistant **Aug 2020-Present** **Mersha Lab** - CCHM Medical Center
 Epigenetics Research **June 2021 - Present**

- Multi Omics integration for asthma research using Deep Learning.
- Identifying subtypes of asthma using CCHMC patient data
Tools: MergeOmics,Diablo, GoM analysis using R and Python
Supervisor: Dr. Tesfaye Mersha

DasGupta Lab - CCHM Medical Center Oncology Department **May 2021 - June 2021**

- Identifying the relationship between Diabetes Obesity and Brain Tumor
- Visual analysis of relationship between sugar consumption and cancer occurrence, the relationship between HbA1c levels and Sugar Consumption, and the relationship between obesity and sugar consumption amongst healthy and Diabetic (Type 1 and 2) patients conducted.
- Insulin, Glucose, HbA1c, Obesity relationship is analyzed amongst Diabetic patients.
- Categorical analysis conducted based on Age, BMI and Country. **Tools:**R, Python, SAS, Jupyter Lab, Pandas,NumPy, Scikitlearn, Seaborn
Supervisors: Dr. Biplab DasGupta, Dr. Abitha Sukumaran

Andorf Lab - CCHM Medical Center Immunology Department **Jan 2021 - Mar 2021**

- Worked on data generated by spectral flow cytometry from blood and placenta tissues.
- Performed tSNE and UMAP to visually check for global differences between the tissues as well as batch effects.
Supervisor: Dr. Sandra Andorf

Aronow Lab - CCHM Medical Center Biomedical Informatics **Sep 2020 - Dec 2020**

- Identified subtypes of Renal Cell Carcinoma using ToppCell and ToppGene tool and differential gene expression.
Supervisor: Dr. Bruce Aronow

Undergraduate Research Assistant

North American University - Houston, TX **Sep 2018 - Aug 2019**

- Accomplished a project Breast Cancer Detection using CNN, achieved 97.36 % accuracy using UCI ML Breast Cancer Wisconsin (Diagnostic) datasets. Github Repo:[Breast Cancer Detection](#)
Supervisor: Dr. Zubeyir Altuntas

Scripps Research Institute - California **Dec 2017 - Feb 2018**

- Solved one of the complex problems of an open-source BiothingsAPI project which is a high-performance biological API for gene annotations.
- Worked on providing a parser for data to integrate PantherDB into the API.
Tools: JSON, PHP, Javascript
Supervisors: Dr. Andrew Su, Dr. Chunlei Wu

HONORS, AWARDS AND ACCOMPLISHMENTS

- University of Cincinnati: Graduate Assistant Scholarship (GAS) - 2022
- University of Cincinnati College of Medicine and CCHMC: Graduate Assistant Scholarship (GAS) - 2020
- University of Cincinnati College of Medicine and CCHMC: University Graduate Assistantship (UGA)- 2020
- Grace Hopper Celebration Student Scholar, 2019
- North American University: Exceptional Merit Scholarship, 2016-2020

- North American University: President's Honor Roll , 2017-2018
- North American University: President's Recognition, 2017-2019
- ACM-ICPC (International Collegiate Programming Contest) Contestant - Baylor University, Waco, TX 2017
Represented North American University at the most prestigious programming contest at Baylor University

LEADERSHIP
EXPERIENCE
VOLUNTEER
ACTIVITIES

- ACM-W Club - Chair, Vice Chair, Senator, Instructor, Lab Leader, Aug 2016-Present
- ACM-ICPC- Co-Coach, Contestant, 2017-2018
- ACM Chapter Data Science Lab - Undergraduate Researcher, Sep 2018- May 2019
- ACM Chapter Bioinformatics Lab - Lab Leader, Researcher, Aug 2016 - Dec 2017
- RAWA Youth Chapter - Youth Magazine Head & Editor, June 2019-Present
- YMCA - Tutor, May 2019
- RAWA Youth Chapter - Outreach & Event Coordinator, June 2018 - June 2019
- Youth Development Center - Tutor, Mentor, Oct 2015 - May 2017
- Young Perspectives Association - Vice President, 2010-2013